

HOMEWORK 3
BIOSTATISTICS 735
DUE SEPTEMBER 30TH, 2026

- **Include your name in the filename of your solutions.** Submit your assignment as a Word file, or a knitted PDF or HTML Quarto file.
- For each applied question, include both code and interpretation of results.
- Be concise but clear: a few well-written sentences with tables/plots is preferred over long text dumps.

1. **(35 points)** The dataset “`parkinsons_updrs.txt`” is composed of a range of biomedical voice measurements from 42 people with early-stage Parkinson’s disease recruited to a six-month trial of a telemonitoring device for remote symptom progression monitoring. The recordings were automatically captured in the patient’s homes. Columns in the table contain subject number, subject age, subject gender, time interval from baseline recruitment date, motor UPDRS, total UPDRS, and 16 biomedical voice measures. The attributes of the data are

- subject number - Integer that uniquely identifies each subject
- age - Subject age
- sex - Subject gender ‘0’ - male, ‘1’ - female
- test_time - Time since recruitment into the trial. The integer part is the number of days since recruitment.
- motor_UPDRS - Clinician’s motor UPDRS score, linearly interpolated
- total_UPDRS - Clinician’s total UPDRS score, linearly interpolated

16 biomedical voice measures

- Jitter(%), Jitter(Abs), Jitter:RAP, Jitter:PPQ5, Jitter:DDP - Several measures of variation in fundamental frequency
- Shimmer, Shimmer(dB), Shimmer:APQ3, Shimmer:APQ5, Shimmer:APQ11, Shimmer:DDA - Several measures of variation in amplitude
- NHR, HNR - Two measures of ratio of noise to tonal components in the voice
- RPDE - A nonlinear dynamical complexity measure
- DFA - Signal fractal scaling exponent
- PPE - A nonlinear measure of fundamental frequency variation

Each row corresponds to one of 5,875 voice recording from these individuals. The main aim of the data is to predict the motor and total UPDRS scores (‘motor_UPDRS’ and ‘total_UPDRS’) from the 16 voice measures.

- (a) (10 pts) Fit a grouped lasso regression predicting **total UPDRS** from the grouped voice predictors. For this question, use "Jitter", "Shimmer", etc. as the groups.
 - (b) (10 pts) For each of the 16 biomedical voice measures, create quadratic and cubic transformations. Fit a grouped lasso with 16 groups, where each group will contain the linear, quadratic, and cubic forms for each of the 16 biomedical voice measures.
 - (c) (5 pts) Fit a regular (non-grouped) lasso models from a, and b.
 - (d) (5 pts) Compare the predictive performance of each of the four models fit in a – c.
 - (e) (5 pts) How does grouped lasso differ from ordinary lasso in interpretation?
2. **(15 points)** This question will use the *Communities and Crime* dataset homework 2.
- (a) (5 pts) Fit a GAM predicting violent crime rate using smooth functions of at least three predictors (choose variables you believe might have nonlinear effects).
 - (b) (5 pts) Plot the smooth terms. Are any relationships clearly nonlinear?
 - (c) (5 pts) Compare predictive performance of the GAM to a linear model with the same predictors. Which has better predictions of the outcome?
3. **(50 points)** The dataset "pima.indians.diabetes3.csv" contains data from 532 females who are at least 21 years of age and of Pima Indian heritage. The dataset consists of:
- `npregnant`: Number of times pregnant
 - `glucose`: Plasma glucose concentration a 2 hours in an oral glucose tolerance test
 - `diastolic.bp`: Diastolic blood pressure (mm Hg)
 - `skinfold.thickness`: Triceps skin fold thickness (mm)
 - `bmi`: Body mass index (weight in kg/(height in m)²)
 - `pedigree`: Diabetes pedigree function
 - `age`: Age (years)
 - `classdigit`: Numerical diabetes indicator
 - `class`: Alphanumeric diabetes variable

The purpose of this study was to determine whether the patient exhibited *any* signs of diabetes mellitus according to World Health Organization criteria (i.e., if the 2-hour post-load plasma glucose level was at least 200 mg/dL at any survey examination or if it was detected during routine care). The two class variables indicate diabetes status using a blood test of 2-hour serum insulin. The goal is to use the other variables to predict diabetes status, as obtaining information on them is easier for patients, less invasive, and less expensive than a blood test. Use all other variables in each analysis.

You are to perform an analysis of this data. In terms of models, you should use at least 2 logistic regression based models (e.g., regular logistic regression, penalized logistic regression, etc.) and at least 2 non-logistic regression based (e.g., Naive Bayes, CART). Your analysis should demonstrate which candidate model has the *best predictive performance*, and provide an unbiased estimate of the best models predictive performance.

In terms of *best predictive performance*, consider the study goals in choosing the most appropriate predictive accuracy metric(s).